Dr. Babasaheb Ambedkar Technological University, Lonere

Summer Examination May 2015

B.Tech Course in Electrical Engineering

Subject: High Voltage Engineering

Semester: VIII

Date: 20/5/2015 **Time: 3 Hours** **Max. Marks: 70**

Instructions to the students:

1) Question No. 1 is compulsory and carry 10 marks whereas Question No. 2 to Question No. 7 carries 12 Marks each.
2) Attempt any five Question from Question No. 2 to Question No. 7
3) Illustrates your answers with neat sketches, diagram, etc. wherever necessary.
4) Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is a part of examination.
5) If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q.1) **Fill in the blanks with correct answers.** (10)

i.) Fault location in an HV cable is done by-----------------------.

ii.) The salt-fog test dine on insulators is -----------------------------------test.

iii.) The peak value of lightning stroke currents are of the order---- ------------.

iv.) Sphere gap measurement is linear and valid for gap spacing less than or equal to ----------------.

v.) A 16-stage impulse voltage generator has stage capacitance of 0.125 μF and a charging voltage of 200 kV. The energy rating in kJ is -----------------------------.

Q.2) **Attempt Any Two of the following.** (12)

a.) Explain the streamer theory of breakdown in gases.

b.) Discuss the various mechanisms of vacuum breakdown.

c.) What will be the breakdown voltage of a spark gap in a gas at pressure = 760 torr at 25° C if A = 15/cm, B = 360/cm, d = 1 mm and $\gamma = 1.5 \times 10^{-4}$?

Q.3) **Solve Any One from the following.** (12)

a.) Discuss in details the breakdown of solid dielectrics in practice.

b.) Explain the following properties of composite dielectrics, (I) effects of multiple layers, (II) effects of layer thickness, and (III) effects of interfaces in detail.

Q.4) **Answer the following questions. (Any Two)** (12)

a.) A Cockcroft - Walton type voltage multiplier has eight stages with capacitances, all equal to 0.05 μF. The supply transformer secondary voltage is 125 kV at a frequency of 150 Hz.

If the load current to be supplied is 5 mA. Find (I) the percentage ripple, (II) the regulation, and (III) number of stages for minimum regulation or voltage drop.

b.) Explain the generation of high frequency ac high voltage method in details.

c.) Explain the basic principle and construction of electrostatic machine Van de Graaff generator for generating high D.C. voltages.

Q.5) **Attempt the following questions.**

a.) What are the factors that affect the sparkover voltage measurements of a sphere gap method? (06)

b.) Explain the principle and construction of an electrostatic voltmeter for very high voltages. What are its merits and demerits for high voltage ac measurements? (06)

Q.6) **Answer the following questions.**

a.) Explain the different sources of origin of switching surges along with typical wave shapes of the switching surges. (06)

b.) Explain the phenomenon Attenuation and distortation of travelling waves in details. (06)

Q.7) **Solve the following questions.**

a.) What are the different high voltage test conducted on the bushings explain in details (06)

b.) Discuss the different test carried on the cables in brief. (06)

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